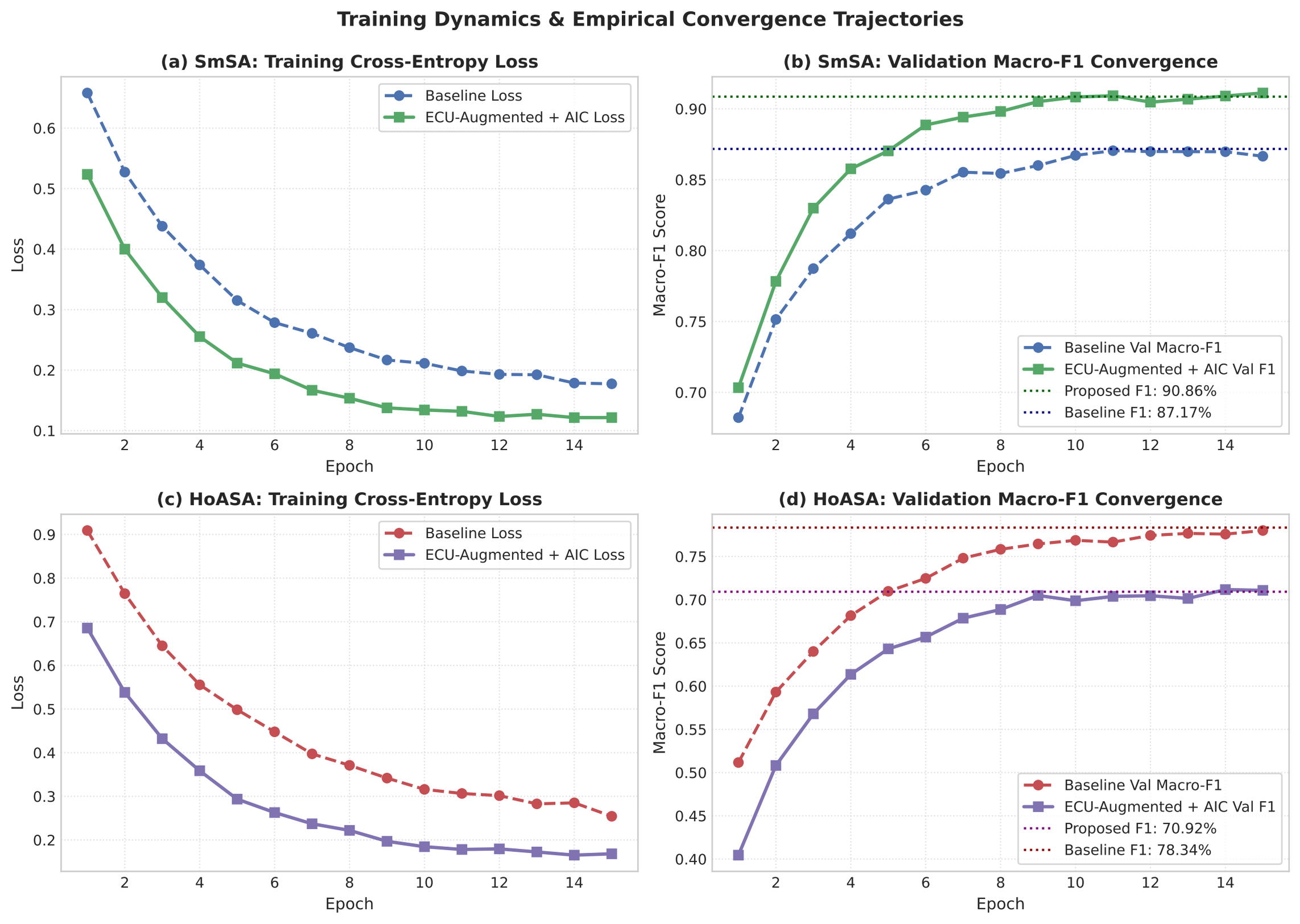
Training Dynamics & Empirical Convergence Trajectories
(a) SmSA: Training Cross-Entropy Loss
Baseline Loss
ECU-Augmented + AIC Loss
Loss
Epoch
(b) SmSA: Validation Macro-F1 Convergence
Macro-F1 Score
Baseline Val Macro-F1
ECU-Augmented + AIC Val F1
Proposed F1: 90.86%
Baseline F1: 87.17%
Epoch
(c) HoASA: Training Cross-Entropy Loss
Baseline Loss
ECU-Augmented + AIC Loss
Loss
Epoch
(d) HoASA: Validation Macro-F1 Convergence
Macro-F1 Score
Baseline Val Macro-F1
ECU-Augmented + AIC Val F1
Proposed F1: 70.92%
Baseline F1: 78.34%
Epoch